

# Krishnaa Vadivel | Software Engineer | Scientific Computing | HPC Workflows

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## Profile

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Software engineer and PhD researcher focused on scientific computing, distributed workflows, numerical software, and engineering tooling. Strong background in Python and C++ development for simulation, automation, data processing, visualization, and infrastructure spanning research and production-adjacent environments.

## Research Experience

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### Yale University, Electrical Engineering

*Research Assistant / PhD Researcher*

*2021–present*

- Develop scientific software for first-principles and many-body workflows in materials research, combining Python and C++ with MPI, OpenMP, PETSc, SLEPc, and Linux-based automation.
- Build distributed and accelerator-enabled workflows for HPC environments including Frontier, Frontera, and Perlmutter, with experience spanning CUDA-, ROCm-, and container-oriented development patterns.
- Create reusable tooling, notes, and structured workflows that bridge theory, implementation, and reproducibility in computational science.
- Extend into ML-assisted and retrieval-driven automation where useful, particularly for scientific-data generation and research-workflow setup.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS talks in 2024, 2025, and 2026 on self-trapped exciton formation and related first-principles work.

## Professional Experience

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### Probe Technologies Inc. / Weatherford Wireline Services

*Software and Embedded Systems Engineer*

*2019–2021*

- Built CUDA-accelerated processing and custom OpenGL visualization software for large acoustic waveform datasets used in engineering diagnostics and field-tool analysis.
- Developed ASP-style services and Microsoft SQL-backed workflows for internal software and field-data management.
- Worked close to embedded and instrumentation systems, giving practical experience with software operating alongside deployed hardware and field constraints.
- Participated in broader engineering support spanning embedded-adjacent software, validation, and integration contexts.

### FindLight

*Photonics Technical Writer*

*2018–2019*

- Produced technical content for optics and photonics topics, demonstrating strong technical communication across specialized engineering subject matter.

## Selected Projects

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### DOLFINx Semiconductor and PDE Workflow

- Built Python workflows around DOLFINx and PETSc for Poisson and drift-diffusion style simulation, including theory notes, implementation scaffolding, and numerical experimentation.

### Agentic Scientific Workflow Generation

- Developed retrieval-driven tooling that reads documentation and academic papers to generate structured starter inputs and workflow artifacts for computational-science tasks.

### Scientific Visualization and GPU Tooling

- Built CUDA/OpenGL-based tooling and related utilities for large numerical datasets, interactive inspection, and engineering communication.

### Photonics and Simulation Code Artifacts

- Created clean, inspectable software artifacts for FDTD-based photonics simulation and related numerical-science workflows.

## Education

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### Yale University

*PhD, Electrical Engineering*

2021–present

Ongoing work combining scientific computing, high-performance workflows, and computational materials research.

### Yale University

*MPhil, Electrical Engineering*

2023

### Yale University

*MS, Electrical Engineering*

2023

### Cornell University

*MEng, Electrical and Computer Engineering*

2017–2018

### Cornell University

*BS, Electrical and Computer Engineering*

2013–2017

## Selected Coursework

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**Computing Foundations:** Theoretical Computer Science; Probabilistic Machine Learning

**Scientific Foundations:** Graduate Quantum Mechanics; Solid State Physics; Many-Body Quantum Physics

**Engineering Foundations:** Signals and systems, embedded systems, semiconductor physics, and numerical/scientific computing practice

## Technical Skills

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**Languages:** Python, C, C++, JavaScript, Bash, PowerShell, SQL

**Scientific Computing:** MPI, OpenMP, PETSc, SLEPc, mpi4py, petsc4py, slepc4py, NumPy, pandas, SciPy, SymPy, matplotlib

**Systems:** Linux command line, CMake, Docker, Docker Compose, Kubernetes, gcloud

**Platforms:** CUDA, OpenGL, Frontier, Frontera, Perlmutter

**ML / Automation:** PyTorch, scikit-learn, agentic workflows, retrieval-driven automation